Supply Chain Risk, Trade and Economic Fragility

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Motivation

- Supply-chain risks affect the attractiveness of sourcing from different countries
  - Natural disasters, geopolitical tensions, and uncertainty about trade policy
- Firms weigh production costs against supply-chain risk when making sourcing decisions
  - Diversifying suppliers reduces risk, may raise production costs
- In a production chain, value of reliability can differ with the position of a good
  - Risk interacts with productivity and substitution possibilities to shape **countries' specialization across stages**
- **Question:** *How does supply-chain risk reshape the global allocation of production and the patterns of specialization?*

This paper

1. **Theory:**
   - General equilibrium model of sequential production under risk
   - Supply-chain links formed ex-ante: choose expected input quantities from origins for stages
   - Input delivery is subject to bilateral, stage-specific shocks, with variance $\sigma^2_{ij,s}$
   - Analytical (exact) characterization of changes in optimal sourcing shares when $\sigma^2_{ij,s}$ changes
   - Analytical results: welfare and risk, risk and comparative advantage, position of risk in supply chain
2. **Quantification**
   - Estimate model with 3 sequential stages
   - Counterfactuals:
     - 2.1 Deterministic benchmark: comparative advantage steepens
     - 2.2 Increases in variance of upstream vs downstream stages: downstream risk increases larger welfare volatility
     - 2.3 Increase in origin-specific risk (China), US trade costs/variance increase
     - 2.4 Increased complexity of the supply chain – riskier regions relatively benefit, more opportunities to diversify

Framework with Sequential Production under Risk

Environment

J countries, S stages; region i endowed with L_i workers

Shocks $\boldsymbol{\chi} = \{\chi_{ji,s}\}$ (mean one) hit **delivered** inputs; **ex-ante** choices not indexed by $\boldsymbol{\chi}$

Households (region i)

Expected log utility over non-traded final output

$$U_i = \mathbb{E}_{\boldsymbol{\chi}}[\log q_i(\boldsymbol{\chi})], \qquad q_i(\boldsymbol{\chi}) = \left(\ell_i^F\right)^{\beta} M_i(\boldsymbol{\chi})^{1-\beta}$$

ℓ_i^F chosen ex ante $\Rightarrow$

$$U_i = \beta \log \ell_i^F + (1-\beta)\,\mathbb{E}_{\boldsymbol{\chi}}[\log M_i(\boldsymbol{\chi})]$$

Technology & trade

Supply-chain aggregate, CRS in delivered inputs & processing labor

$$M_i(\boldsymbol{\chi}) = f_i\left(\{\chi_{ji,s} M_{ji,s}\}_{j,s}, \{\ell_{i,s}^p\}_{s\geq 1}\right)$$

$M_{ji,s}$: ex-ante input quantity/link choice $j \to i$, stage s; delivered $\chi_{ji,s} M_{ji,s}$

One unit of stage-s input $j \to i$ uses $\tau_{ji,s}/a_{j,s}$ of j-labor

Ex-ante labor resource constraint (each origin j)

$$\ell_j^F + \sum_i \sum_s \frac{\tau_{ji,s} M_{ji,s}}{a_{j,s}} + \sum_s \ell_{j,s}^p \;\leq\; L_j$$

Planner's problem

Υ_i: welfare weight; Ξ_j: shadow value of country j's labor constraint.

$$\max_{\{\ell_i^F\},\{M_{ji,s}\},\{\ell_{i,s}^p\}} \sum_i \Upsilon_i \left[\beta \log \ell_i^F + (1-\beta)\, \mathbb{E}_{\boldsymbol{\chi}} \log M_i(\boldsymbol{\chi})\right]$$

$$\text{s.t.} \quad \text{resource constraint } (\Xi_j)\ \forall j$$

FOCs final labor $\Xi_i = \Upsilon_i \beta / \ell_i^F$; each input's expected marginal contribution to welfare equals its shadow marginal resource cost.

$$\Xi_j \frac{\tau_{ji,s} M_{ji,s}}{a_{j,s}} = (1-\beta)\Upsilon_i\, \mathbb{E}_{\boldsymbol{\chi}}\Big[\underbrace{\mathcal{E}_{ji,s}(\boldsymbol{\chi})}_{\substack{\text{Pointwise output} \\ \text{elasticity w.r.t. } M_{ji,s}}} \Big], \qquad \Xi_i\, \ell_{i,s}^p = (1-\beta)\Upsilon_i\, \mathbb{E}_{\boldsymbol{\chi}}\Big[\underbrace{\mathcal{E}_{i,s}^p(\boldsymbol{\chi})}_{\substack{\text{Pointwise output} \\ \text{elasticity w.r.t. } \ell_{i,s}^p}} \Big]$$

CRS $\Rightarrow \sum_{j,s} \mathcal{E}_{ji,s} + \sum_s \mathcal{E}_{i,s}^p = 1$ state by state $\Rightarrow$ adding-up:

$$\sum_{j,s} \Xi_j \frac{\tau_{ji,s} M_{ji,s}}{a_{j,s}} + \sum_s \Xi_i\, \ell_{i,s}^p = (1-\beta)\Upsilon_i$$

Each input's shadow-cost share in i's composite $=$ its expected output elasticity

Decentralization

Under balanced trade, the competitive allocation solves the planner's problem for appropriately chosen Pareto weights.

- Competitive eq.: MC pricing $p_{ji,s} = \tau_{ji,s} w_j / a_{j,s}$, labor markets clear, trade balance
- Competitive equilibrium implies $\ell_i^F = \beta L_i$. With no-transfer weights $\Upsilon_i = w_i L_i$, the planner's final-labor FOC gives $\Xi_i = w_i$.
- Substitute into the planner link FOC $\Rightarrow$ ex-ante optimality

$$p_{ji,s} M_{ji,s} = (1-\beta)\, w_i L_i\, \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}(\boldsymbol{\chi})]$$

Expenditure share = expected output elasticity

$$s_{ji,s} \equiv \frac{p_{ji,s} M_{ji,s}}{(1-\beta) w_i L_i} = \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}(\boldsymbol{\chi})]\,, \qquad s^p_{i,s} \equiv \frac{w_i \ell^p_{i,s}}{(1-\beta) w_i L_i} = \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}^p_{i,s}], \qquad \sum_{j,s} s_{ji,s} + \sum_s s^p_{i,s} = 1$$

- Observed expenditure shares are a **sufficient statistic for expected output elasticities**.

Share definitions

CRRA shares

Sequential production

New inputs at each stage

$$B_{i,s} = \left[\sum_{j=1}^{J} (\chi_{ji,s} M_{ji,s})^{\rho_s}\right]^{1/\rho_s}, \qquad s = 1, \ldots, S.$$

Inherited output and processing labor

$$\begin{aligned} y_{i,1} &= B_{i,1}, \\ X^p_{i,s} &:= y^{\psi_s}_{i,s-1} (\ell^p_{i,s})^{1-\psi_s}, \\ y_{i,s} &= \left[\alpha_{i,s} (X^p_{i,s})^{\eta} + B^{\eta}_{i,s}\right]^{1/\eta}, \quad s = 2, \ldots, S, \\ M_i &= y_{i,S}. \end{aligned}$$

Realized production elasticities

Origin's contribution within a stage

$$\begin{aligned} \omega_{ji,s}(\boldsymbol{\chi}) &= \frac{\partial \ln B_{i,s}}{\partial \ln(\chi_{ji,s} M_{ji,s})} \\ &= \frac{(\chi_{ji,s} M_{ji,s})^{\rho_s}}{\sum_{j'=1}^{J} (\chi_{j'i,s} M_{j'i,s})^{\rho_s}}. \end{aligned}$$

New inputs' contribution to stage output

$$\lambda_{i,s}(\boldsymbol{\chi}) = \frac{\partial \ln y_{i,s}}{\partial \ln B_{i,s}} = \frac{B^{\eta}_{i,s}}{\alpha_{i,s} (X^p_{i,s})^{\eta} + B^{\eta}_{i,s}},$$

for $s \geq 2$; at stage 1, $\lambda_{i,1}(\boldsymbol{\chi}) = 1$.

Which share? Roundabout equivalence General IO networks

Pointwise elasticity: propagation along the chain

Effect of one input on the completed intermediate composite

$$\underbrace{\mathcal{E}_{ji,s}(\boldsymbol{\chi})}_{\partial \ln M_i / \partial \ln M_{ji,s}} = \omega_{ji,s}(\boldsymbol{\chi})\, \lambda_{i,s}(\boldsymbol{\chi}) \prod_{r=s+1}^{S} \big[\psi_r(1-\lambda_{i,r}(\boldsymbol{\chi}))\big].$$

- $\omega_{ji,s}$: how strongly the stage-s input bundle $B_{i,s}$ responds to deliveries from origin j.
- $\lambda_{i,s}$: how strongly stage-s output responds to its new input bundle $B_{i,s}$; $\lambda_{i,1} = 1$.
- $\prod_{r=s+1}^{S}$: multiply the downstream elasticities $\psi_r(1-\lambda_{i,r})$ across all later stages r.

Special cases

- **Cobb–Douglas:** ω and λ are constants, so $\mathcal{E}_{ji,s}$ does not vary across states.
- **Leontief ($\eta \to -\infty$):** $\lambda_{i,s}(\boldsymbol{\chi}) = \mathbf{1}$ if $B_{i,s}$ smaller than inherited work-in-progress
  - Input contributes if its stage is tightest constraint
  - $s_{ji,s}$ is probability that s is binding constraint $\times$ the CES within-stage share $\times$ processed stage share surviving downstream

Elasticities vs spending shares | Cobb–Douglas IO networks

Optimal sourcing and welfare

Ex-ante trade value: $X_{ji,s} := p_{ji,s} M_{ji,s}$, with $p_{ji,s} = \tau_{ji,s} w_j / a_{j,s}$.

$$s_{ji,s} = \frac{X_{ji,s}}{(1-\beta) w_i L_i} = \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}(\boldsymbol{\chi})].$$

The endogenous risk-and-technology term

$$\Psi_{ji,s} := \mathbb{E}_{\boldsymbol{\chi}}\left[\frac{\chi_{ji,s}^{\rho_s}}{\sum_{j'=1}^{J} (\chi_{j'i,s} M_{j'i,s})^{\rho_s}} \lambda_{i,s}(\boldsymbol{\chi}) \prod_{r=s+1}^{S} \left[\psi_r (1 - \lambda_{i,r}(\boldsymbol{\chi})) \right] \right].$$

Gravity-like expenditure equation

$$X_{ji,s} = p_{ji,s}^{-\rho_s/(1-\rho_s)} \left[(1-\beta) w_i L_i \Psi_{ji,s} \right]^{1/(1-\rho_s)}.$$

$\Psi_{ji,s}$ depends on all orders and their joint risk: this is an implicit sourcing condition.

Welfare: expected log consumption, $\mathcal{W}_i = U_i$,

$$\mathcal{W}_i = \beta \ln(\beta L_i) + (1-\beta) \mathbb{E}_{\boldsymbol{\chi}}[\ln M_i(\boldsymbol{\chi})].$$

General equilibrium

Risk, relative prices, and welfare

Direct welfare effects: holding prices fixed

$$\left.\frac{\partial \mathcal{W}_i}{\partial \sigma^2_{ji,s}}\right|_{w,p} = \frac{1-\beta}{2}\,\mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}(\vartheta_{ji,s}-1)] \le 0, \qquad \left.\frac{\partial \mathcal{W}_i}{\partial \log \tau_{ji,s}}\right|_{w} = -(1-\beta)s_{ji,s} \le 0.$$

Country welfare also depends on relative prices. Increase origin 1's log-shock variances by a common amount: $d\sigma^2_{1i,s} = d\nu_1$ for every destination i and stage s.

$$\frac{d\mathcal{W}_i}{d\nu_1} = \underbrace{\left.\frac{\partial \mathcal{W}_i}{\partial \nu_1}\right|_{w,p}}_{\text{direct risk effect}} + \underbrace{\sum_k \frac{\partial \mathcal{W}_i}{\partial \log w_k}\frac{d\log w_k}{d\nu_1}}_{R_i\text{: wage / terms-of-trade effect}}.$$

Global welfare for a small risk increase: Let $\bar{\Upsilon}_i := \bar{w}_i L_i / \sum_k \bar{w}_k L_k$, then

$$\sum_i \bar{\Upsilon}_i R_i = 0, \qquad \sum_i \bar{\Upsilon}_i \frac{d\mathcal{W}_i}{d\nu_1} = \sum_i \bar{\Upsilon}_i \left.\frac{\partial \mathcal{W}_i}{\partial \nu_1}\right|_{w,p} \le 0.$$

Risk-adjusted comparative advantage

Assume a common within-stage CES power $\rho_s = \rho \in (0,1)$. Trade value is $X_{ji,s} = p_{ji,s} M_{ji,s}$; i is the destination.

Relative exports across origins and stages (following CDK, 2012)

$$\Delta^{s,s'}_{jj',i} := \ln \frac{X_{j'i,s}}{X_{ji,s}} - \ln \frac{X_{j'i,s'}}{X_{ji,s'}} = \frac{\rho}{1-\rho} \mathrm{CA}^{s,s'}_{jj',i} + W^{s,s'}_{jj',i}.$$

$$\mathrm{CA}^{s,s'}_{jj',i} := \ln \frac{a_{j',s} \tau_{ji,s}}{a_{j,s} \tau_{j'i,s}} - \ln \frac{a_{j',s'} \tau_{ji,s'}}{a_{j,s'} \tau_{j'i,s'}},$$

$$W^{s,s'}_{jj',i} := \frac{1}{1-\rho} \left[\ln \frac{\Psi_{j'i,s}}{\Psi_{ji,s}} - \ln \frac{\Psi_{j'i,s'}}{\Psi_{ji,s'}} \right].$$

Risk-adjusted productivity: $\widetilde{a}_{ji,s} := (a_{j,s}/\tau_{ji,s}) \Psi^{1/\rho}_{ji,s}$. Then

$$\Delta^{s,s'}_{jj',i} = \frac{\rho}{1-\rho} \left[\ln \frac{\widetilde{a}_{j'i,s}}{\widetilde{a}_{ji,s}} - \ln \frac{\widetilde{a}_{j'i,s'}}{\widetilde{a}_{ji,s'}} \right].$$

- CA: productivity adjusted for trade costs; W: the endogenous Ψ terms.
- Within-stage diversification and cross-stage complementarity shape Ψ and can attenuate observed comparative advantage.

In paper: some conditions for attenuation, amplified by cascade effects

Risk changes sourcing through two channels

- For each origin j and stage s, let $N^M_{j,s}$ be order-production labor
- Consider a common origin-risk increase, $d\sigma^2_{ji,r} = d\nu_j$, from the deterministic benchmark

Origin Specialization

$$\frac{d}{d\nu_j} \ln \frac{N^M_{j,S}}{N^M_{j,1}}\bigg|_w = \underbrace{-\sum_i m^0_{i|j,S}(1-\omega^0_{ji,S})F_{i,S} + \sum_i m^0_{i|j,1}(1-\omega^0_{ji,1})F_{i,1}}_{\text{origin reallocation within stages}} + \underbrace{\sum_i m^0_{i|j,S}\frac{d\ln T_{i,S}}{d\nu_j} - \sum_i m^0_{i|j,1}\frac{d\ln T_{i,1}}{d\nu_j}}_{\text{stage-expenditure shift}}.$$

where $m^0_{i|j,s}$ is the stage-s sales weights of origin j in the deterministic benchmark. Three margins:

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1. Origin reallocation within stages

$$\frac{d\ln\left(s_{ji,s}/s_{\ell i,s}\right)}{d\nu_j} = \underbrace{-\frac{\rho_s}{2(1-\rho_s)}\left[1-\rho_s\left(1-2\omega^0_{ji,s}\right)\right]}_{\substack{\text{within-nest}\\ \text{CES reallocation}}} \underbrace{-\frac{\rho_s}{1-\rho_s}\,\omega^0_{ji,s}\left[-\eta(1-\Lambda_{i,s})\right]}_{\text{accumulated-value term}} =: -F_{i,s}$$

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2. Stage expenditure shift

$$\frac{d\ln s_{ji,s}}{d\nu_j} = -(1-\omega^0_{ji,s})F_{i,s} + \frac{d\ln T_{i,s}}{d\nu_j}, \quad T_{i,s} := \sum_\ell s_{\ell i,s}$$

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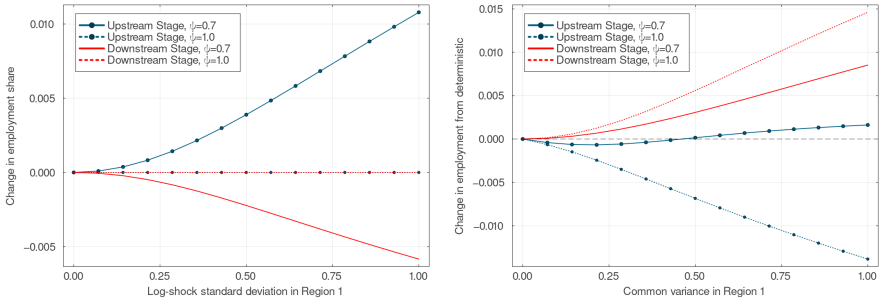
$$\frac{d\ln s_{ji,s}}{d\nu_j} = -(1-\omega^0_{ji,s})F_{i,s} + \frac{d\ln T_{i,s}}{d\nu_j}, \quad T_{i,s} := \sum_\ell s_{\ell i,s}$$

3. GE-effect

Risk and specialization: position and substitution

- Two countries, three stages: $a = \tau = \alpha = 1$, $\eta = -1$; compare $\psi_s = 0.7$ and $\psi_s = 1$.
- Raise risk at all stages in country 1; country 2 remains riskless.

Figure: Changes in country 1's production-and-processing employment shares



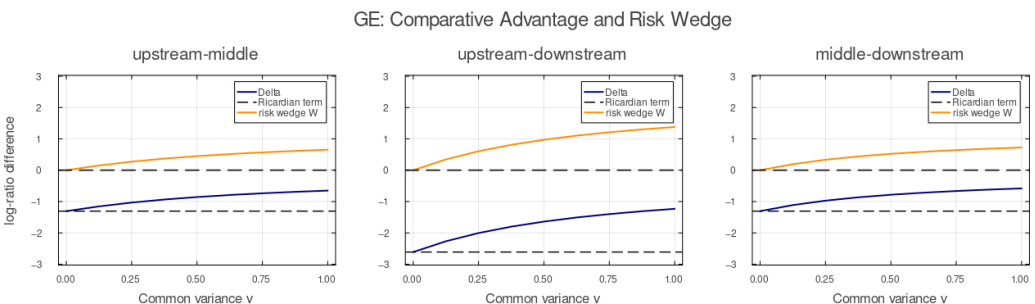
(a) Common $\rho_s = 0.7$ (b) Different ρ_s: $\rho_1 > \rho_2 > \rho_3$

Horizontal axis: log-shock standard deviation σ; changes are relative to each economy's deterministic benchmark.

Interpreting the comparison

More risk attenuates revealed comparative advantage

- $J = 2$, $S = 3$, $\eta = -1$, $\psi_s = 0.7$: opposing productivity advantages and common risk:
  - Region 1: $a = (1.4, 1.0, 0.8)$, $\alpha_2 = \alpha_3 = 2/3$
  - Region 2: $a = (0.8, 1.0, 1.4)$, $\alpha_2 = \alpha_3 = 2/3$
- Productivity differences are fixed as risk rises.



Horizontal axis: common variance σ^2 of log delivery shocks, raised in both countries and all stages.

Quantification

Data and estimation strategy

- **Data:** 50 countries and three industry groups ordered by upstreamness (Antràs et al., 2012). Trade coverage: 1995–2023; common estimation panel: 1995–2021.
- **Objects to estimate:** within-stage CES powers ρ_s, cross-stage power η, origin-stage competitiveness, bilateral trade costs, and the joint shock process.

Challenge: gravity without IIA

The model does not satisfy independence of irrelevant alternatives (IIA): relative sourcing between two origins depends on other suppliers. Its gravity equation includes a bilateral-stage-year term $\Psi_{ji,s,t}$, which we construct within the model.

1. Estimate within-stage substitution and trade costs with Poisson pseudo-maximum likelihood (PPML), updating $\Psi_{ji,s,t}$ using the model. Here t indexes years.
2. For each candidate η, fit destination-stage weights to spending-share levels. Choose η to match how relative stage spending responds to tariff-induced cost changes over time.
3. Use structural trade residuals to estimate log-shock standard deviations and correlations.

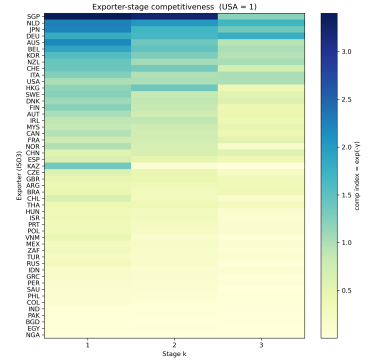
Estimation details | Conditional precision | Stage groups and IO links

Which variation identifies each parameter?

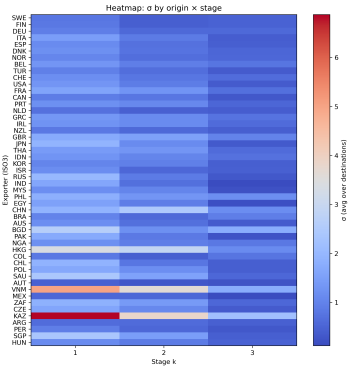
- ρ_s: Responses of bilateral trade to *within-stage* tariff changes, controlling for origin-stage-year, destination-stage-year, and bilateral-stage fixed effects and the model's risk term.
- $a_{j,s}$: Persistent origin-stage differences in trade, recovered under the model's trade-cost normalizations and controlling for wages and the risk term.
- **Gravity coefficients:** Relate persistent bilateral trade wedges to distance, common language, and border effects, controlling for origin and destination effects.
- $\alpha_{i,s}$: Destination-stage branch coefficients match average observed stage shares $S^{\mathrm{D}}_{i,s}$, normalized over purchased inputs
- η: Match the response of relative stage expenditure to tariff-induced cost changes. Stage weights fit share levels; η governs the response over time.
- **Log-shock s.d. $\sigma_{j,s}$ & correlations $R^J \otimes R^S$:** Dispersion and co-movement of structural trade residuals. Country and stage correlation matrices R^J and R^S impose a separable correlation structure. Details

Covariance restriction Lognormal tails

Estimated substitution, competitiveness, and risk



(a) Origin-stage competitiveness



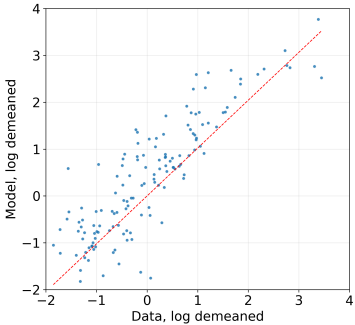
(b) Log-shock s.d. $\sigma_{j,s}$

Stage	ρ_s
1	**0.87**
Industry:	*Iron & steel, Mining (energy), Mining (non-energy), Wood, Paper,Rubber & plastics*
2	**0.84**
Industry:	*Electrical machin., Fabricated metal, Coke & petrol. Prod., Chemicals, Non-metallic prod., Agri., Computer, medical equip.*
3	**0.68**
Industry:	*Ships & boats, Machinery equip., Textile, Other manuf., Motor vehicle, F&B*

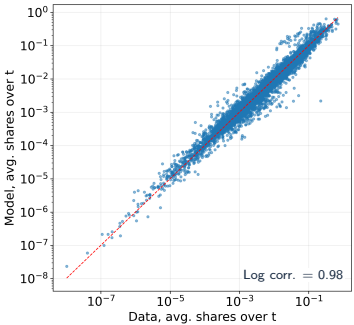
(c) Within-stage CES power ρ_s

- Estimated $\eta = -0.74$, cross-stage substitution elasticity $1/(1-\eta) \approx 0.57$ Shock interpretation

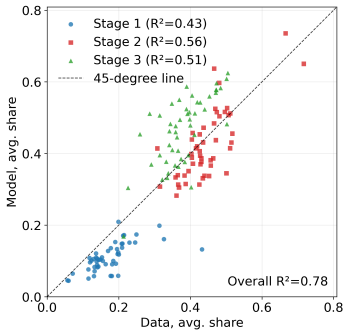
Model fit: trade flows and spending shares



(a) Origin-stage exports, log demeaned



(b) Mean foreign-sourcing shares

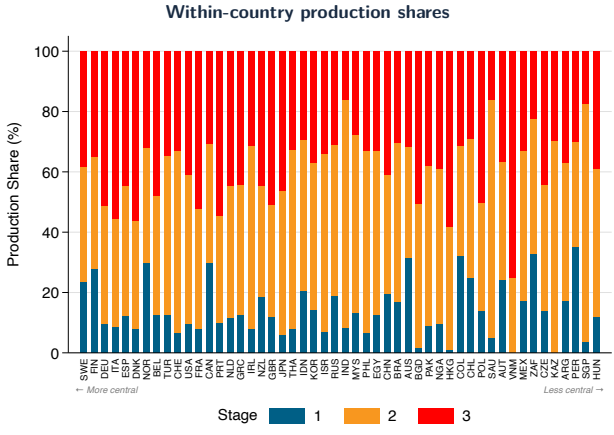


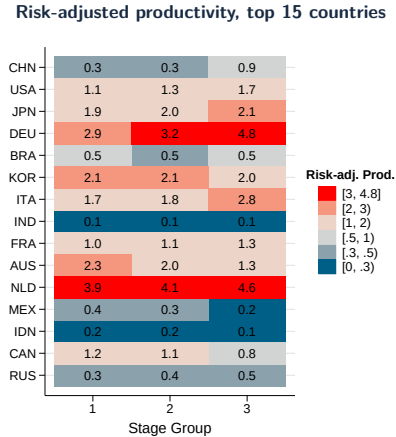
(c) Mean stage expenditure shares

In-sample fit using estimated bilateral-stage wedges; sourcing-share panel reports log correlation.

Identification and precision

Baseline economy: specialization and productivity



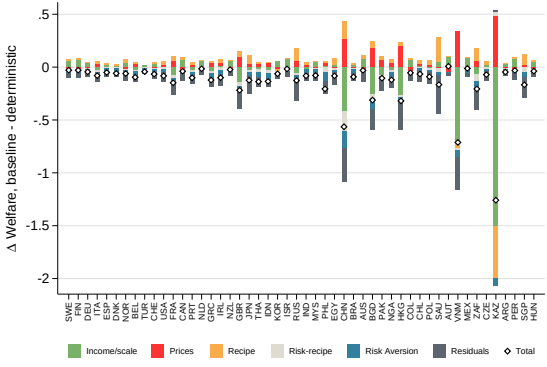


- More downstream, more productive, less risky

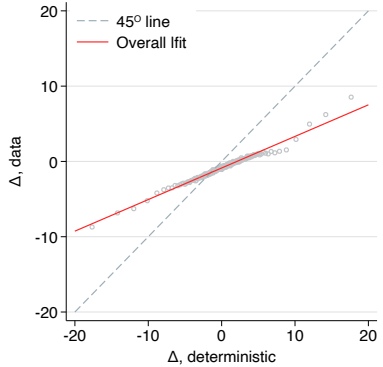
Risk-adjusted productivity | Centrality

Counterfactuals

Risk lowers aggregate welfare and attenuates specialization



(a) Selected welfare contributions and total (ΔU_i)

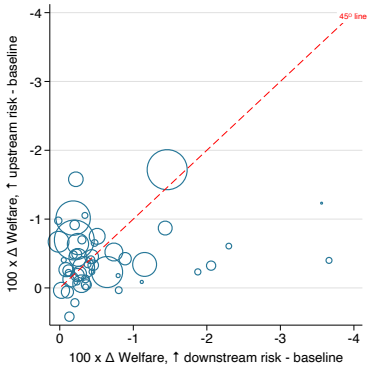


(b) Log-export double differences: data vs. deterministic model

- Observed specialization is weaker than in the deterministic benchmark
  - The right panel compares log export ratios across two origins and two stages, for a given destination

Global welfare and consumption equivalents

Where risk occurs affects output volatility



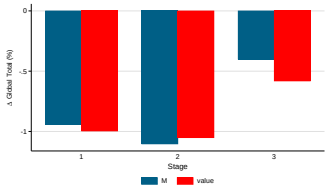
(a) Welfare change (log points); bubble size: income



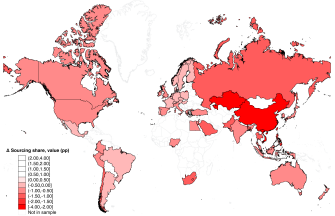
(b) Composite-output CV: downstream-risk / upstream-risk

- Raise risk in stage 1 or stage 3 to increase trade-weighted log-shock variance by 5%, using baseline weights Risk measure
- Downstream risk produces a higher $\mathrm{CV}(M_i) = \mathrm{sd}(M_i)/\mathbb{E}[M_i]$ in 45 of 50 countries

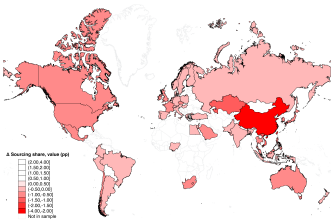
Higher Chinese risk shifts sourcing across stages



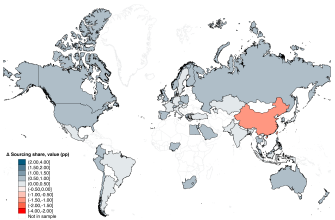
(a) World input production: percent change



(b) Sourcing from China: stage 1 (pp)



(c) Sourcing from China: stage 2 (pp)



(d) Sourcing from China: stage 3 (pp)

- Raise China's risk at all stages to increase aggregate log-shock variance by 5% Risk measure
- World input quantities fall at every stage
- On average across foreign buyers, China loses upstream sourcing share and gains downstream share
  - Upstream inputs are more substitutable across origins

Specialization mechanism

Higher Chinese risk: welfare and input prices

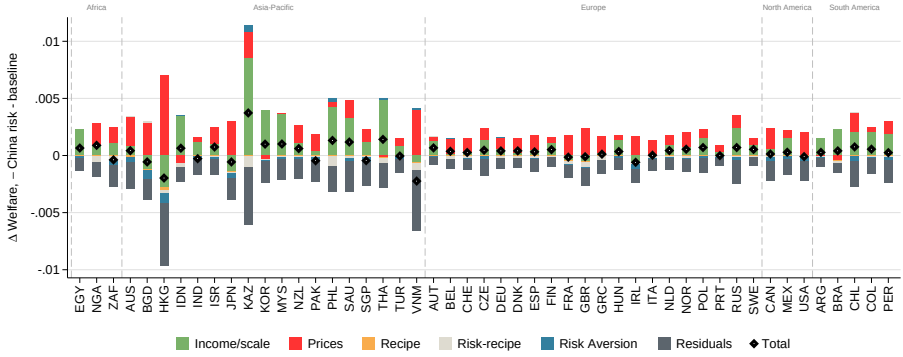


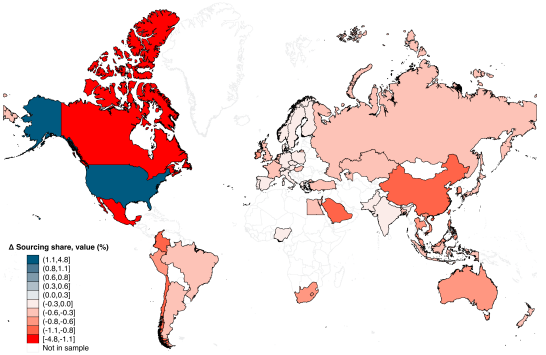
Figure: Welfare outside China: selected contributions and total (ΔU_i)

- Prices of inputs produced in China fall by 8.0%; China's consumption-equivalent welfare falls by 3.88%

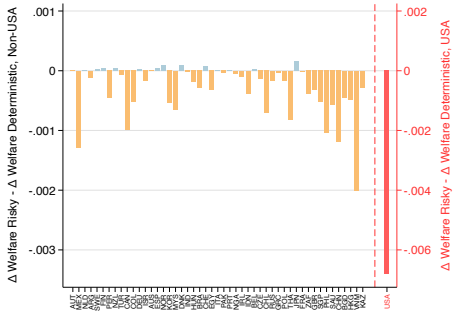
Prices vs welfare contributions

Alternative: US

Higher US import costs change sourcing and welfare



(a) Change in sourcing from the US (percentage points)

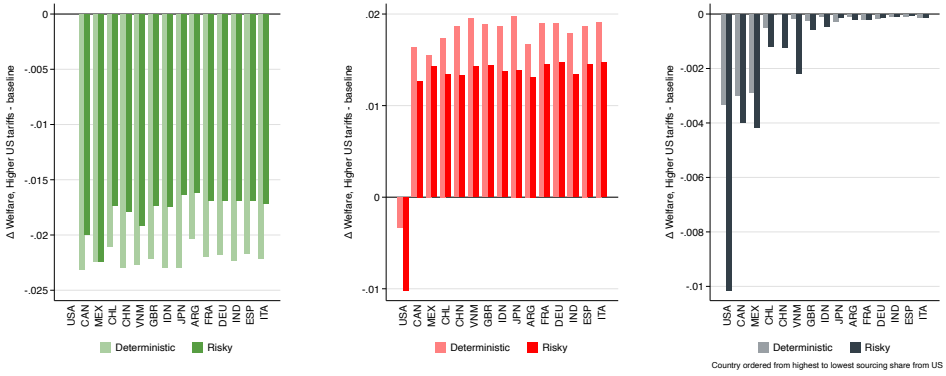


(b) Welfare effect: risky minus deterministic (log units)

- Add 0.175 to iceberg costs on foreign inputs imported by the US; domestic costs stay fixed
- US buyers source more domestically; foreign buyers generally source less from the US By stage
- Welfare effects differ across countries and between risky and deterministic economies

US import costs: income and input-price channels

Figure: Welfare effects relative to each environment's baseline (log units), selected countries



(a) Income/scale (b) Bought-input prices (c) Total

- The US sources less domestically under risk, both before and after the cost increase
- Greater reliance on US inputs under risk weakens the price offset, raising welfare losses in most countries.

All countries | Income | Input prices | US domestic share

Three versus nine stages: additional diversification opportunities

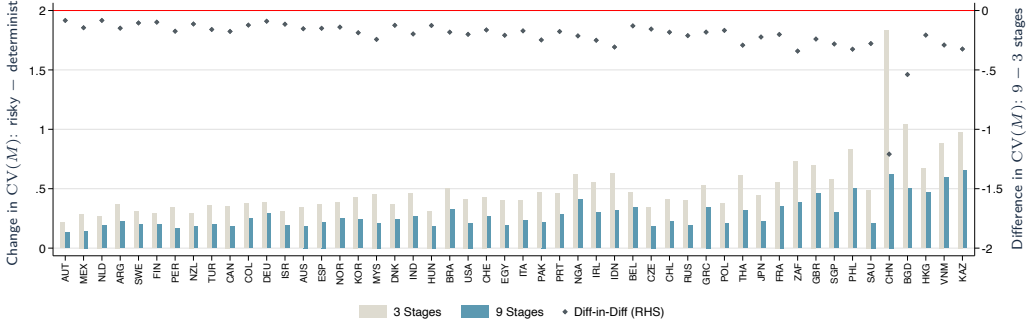


Figure: Composite-output volatility with three and nine stages

- Replace each of the three stages with three copies; log delivery shocks within each triplet have correlation 0.5
- Bars report $\mathrm{CV}(M_i)$ for each chain length; diamonds show nine-stage minus three-stage CV
  - Composite-output CV falls in all 50 countries; the experiment changes both chain length and diversification opportunities

Conclusion

- A quantitative model of production chains with input quantities and labor chosen before delivery shocks
- Risk changes sourcing within stages, spending across stages, and equilibrium wages
- Counterfactuals:
  - Delivery risk weakens revealed comparative advantage relative to a deterministic economy
  - Downstream risk generates more composite-output volatility than upstream risk in most countries
  - Risk and trade-cost changes have different effects across countries; diversification opportunities matter

Discussion appendix

Appendix

Appendix: mechanisms and quantitative interpretation

Production and preferences

Notation guide | Share definitions

Exact roundabout representation

General domestic IO networks

IO: timing and estimation | Cobb–Douglas

CRRA: derive expenditure shares

Real and nominal SDF

Deterministic nominal income

Evidence and quantitative interpretation

Stage groups and IO links

Eta: conditional SE and profile

Covariance restriction | Shock tails

Welfare: weights and units

China: specialization | Prices

Interpreting comparative statics

Back to conclusion | Estimation procedure

Notation guide: quantities and parameters

Indices and quantities

$j \to i$: origin to destination; $s = 1, \ldots, S$: stage.

r: another production stage; t: year in estimation; J: number of countries.

$M_{ji,s}$: ex-ante bilateral input quantity.

$\chi_{ji,s} M_{ji,s}$: realized delivered input.

$B_{i,s}$: new-input bundle at stage s.

$y_{i,s}$: output after stage s; $M_i = y_{i,S}$.

$\ell_i^F, \ell_{i,s}^p$: final-good and processing labor.

$C_i = q_i$: final consumption; $U_i = \mathcal{W}_i$: expected log consumption in the baseline.

Technology, prices, and risk

$a_{j,s}$: input productivity; $\tau_{ji,s} \geq 1$: iceberg trade cost.

$p_{ji,s} = \tau_{ji,s} w_j / a_{j,s}$: price per ordered unit; $X_{ji,s} = p_{ji,s} M_{ji,s}$: trade value.

β: final-production labor exponent.

ρ_s, η: CES powers, with substitution elasticities $1/(1-\rho_s)$ and $1/(1-\eta)$.

$\alpha_{i,s} > 0$: weight on processed work-in-progress ($s \geq 2$).

ψ_s: inherited-output exponent; $1 - \psi_s$: processing-labor exponent ($s \geq 2$).

σ: standard deviation of $\log \chi$; σ^2: its variance, not $\mathrm{Var}(\chi)$.

State arguments are shown when needed. Superscript 0 in local theory means the deterministic benchmark; empirical counterfactuals start from the estimated risky baseline.

Back to environment · Which share? · Appendix guide

Notation guide: elasticity responses and propagation

How the realized output elasticity responds to a delivery shock

$$\vartheta_{ji,s} := \frac{\partial \ln \mathcal{E}_{ji,s}}{\partial \ln \chi_{ji,s}}, \qquad \zeta_{ji,s} := \frac{\partial \vartheta_{ji,s}}{\partial \ln \chi_{ji,s}}.$$

Holding orders and processing labor fixed, and suppressing the state argument,

$$\begin{aligned}\vartheta_{ji,s} = {} & \rho_s(1-\omega_{ji,s}) + \eta\omega_{ji,s}(1-\lambda_{i,s}) \\ & + \eta\omega_{ji,s}\lambda_{i,s} \sum_{r=s+1}^{S} \psi_r \lambda_{i,r} \prod_{u=s+1}^{r-1} \psi_u(1-\lambda_{i,u}).\end{aligned}$$

Deterministic cascade weight for the specialization result

$$\Lambda_{i,s} := \lambda^0_{i,s}\left[\prod_{r=s+1}^{S} \psi_r(1-\lambda^0_{i,r}) + \sum_{r=s+1}^{S}(1-\psi_r)\prod_{u=s+1}^{r-1}\psi_u(1-\lambda^0_{i,u})\right].$$

- $\lambda_{i,s}$ is the elasticity of stage-s output to its new-input bundle. Uppercase $\Lambda_{i,s}$ also accounts for later stages, evaluated at zero risk (superscript 0); $1-\Lambda_{i,s}$ measures accumulated-value exposure.
- At $s=S$, $\Lambda_{i,S} = \lambda^0_{i,S}$. Empty sums equal 0; empty products equal 1.

Back to risk responses | Back to specialization | Appendix guide

Notation guide: production elasticities and spending shares

1. Realized production elasticities: hold the shock state and other inputs fixed

$$\omega_{ji,s} = \frac{\partial \ln B_{i,s}}{\partial \ln(\chi_{ji,s} M_{ji,s})}, \qquad \lambda_{i,s} = \frac{\partial \ln y_{i,s}}{\partial \ln B_{i,s}}, \qquad \mathcal{E}_{ji,s} = \frac{\partial \ln M_i}{\partial \ln M_{ji,s}}.$$

2. Ex-ante expenditure shares: supply-chain budget $B_i^{\text{tot}} = (1-\beta) w_i L_i$

$$s_{ji,s} := \frac{X_{ji,s}}{B_i^{\text{tot}}}, \qquad s_{i,s}^{p} := \frac{w_i \ell_{i,s}^{p}}{B_i^{\text{tot}}}, \qquad \sum_{j,s} s_{ji,s} + \sum_{s} s_{i,s}^{p} = 1.$$

3. Stage budget and within-stage spending shares

$$T_{i,s} := \sum_{j} s_{ji,s}, \qquad \pi_{ji,s} := \frac{s_{ji,s}}{T_{i,s}} = \frac{X_{ji,s}}{\sum_{j'} X_{j'i,s}}, \qquad \sum_{j} \pi_{ji,s} = 1.$$

From production elasticities to expenditure shares

Under log utility, $s_{ji,s} = \mathbb{E}[\mathcal{E}_{ji,s}]$. In general, $\pi_{ji,s} \neq \omega_{ji,s}(\chi)$. They coincide at the deterministic benchmark: $\pi_{ji,s}^{0} = \omega_{ji,s}^{0}$.

$T_{i,s}$ divides stage-s input spending by total supply-chain spending, including processing labor. The estimation share $S_{i,s} = T_{i,s} / \sum_r T_{i,r}$ instead divides by purchased-input spending only.

Back to point-wise elasticity · Back to expenditure shares · Appendix guide

Stein's Lemma and mean-one delivery risk

Suppress link indices. Write the log delivery shock as

$$v := \log \chi = -\frac{\sigma^2}{2} + \sigma\varepsilon, \qquad \varepsilon \sim N(0,1), \qquad \mathbb{E}[\chi] = 1.$$

Let f be twice continuously differentiable with bounded first and second derivatives. Primes denote derivatives with respect to v.

Mean-one lognormal variance derivative

$$\frac{\partial \mathbb{E}[f(v)]}{\partial \sigma^2} = \frac{1}{2}\,\mathbb{E}\left[f''(v) - f'(v)\right], \qquad \sigma^2 > 0.$$

Derivation. Stein's identity gives $\mathbb{E}[\varepsilon f'(v)] = \sigma \mathbb{E}[f''(v)]$. Thus

$$\frac{\partial \mathbb{E}[f(v)]}{\partial \sigma^2} = \frac{1}{2\sigma}\mathbb{E}\left[f'(v)(\varepsilon - \sigma)\right] = \frac{1}{2}\mathbb{E}\left[f''(v) - f'(v)\right].$$

Mean correction. The $-f'(v)$ term comes from setting the log mean to $-\sigma^2/2$ so that expected deliveries remain one as variance rises.

Application to sourcing. Hold choices fixed and condition on the other independent shocks:

$$f(v) = \mathcal{E}_{ji,s}, \qquad f'(v) = \mathcal{E}_{ji,s}\vartheta_{ji,s}, \qquad f''(v) = \mathcal{E}_{ji,s}(\vartheta_{ji,s}^2 + \zeta_{ji,s}).$$

At $\sigma^2 = 0$, the right derivative is $\frac{1}{2}[f''(0) - f'(0)]$.

Back to risk responses

Appendix guide

How orders respond to risk

Order response: orders adjust; processing labor, wages, and prices fixed. Stack i's orders, cost shares, and output elasticities in $\mathbf{M}_i$, $\mathbf{s}_i$, and $\boldsymbol{\mathcal{E}}_i$. When $\mathbf{I} - \mathbf{A}_i$ is nonsingular,

$$\left.\frac{d\mathbf{s}_i}{d\sigma^2_{ji,s}}\right|_{\boldsymbol{\ell}_i^p, w, p} = \mathbf{D}_i(\mathbf{I} - \mathbf{A}_i)^{-1}\mathbf{D}_i^{-1}\mathbf{h}_i, \qquad \mathbf{D}_i := \frac{\operatorname{diag}\{p_{ki,r}\}_{k,r}}{(1-\beta)w_i L_i}.$$

$$\boldsymbol{\Gamma}_i := \frac{\partial \mathbb{E}[\boldsymbol{\mathcal{E}}_i]}{\partial \mathbf{M}_i}, \qquad \mathbf{A}_i := \mathbf{D}_i^{-1}\boldsymbol{\Gamma}_i, \qquad \mathbf{h}_i := \left.\frac{\partial \mathbb{E}[\boldsymbol{\mathcal{E}}_i]}{\partial \sigma^2_{ji,s}}\right|_{\mathbf{M}_i, \boldsymbol{\ell}_i^p, w, p}.$$

Back to risk responses

Appendix guide

Roundabout production can preserve the net technology

Let $G > 0$ be the output of the original stage technology after deliveries. $\mu \in (0,1)$ is a **fixed technology parameter**; R is an input of the same good as gross output Q:

$$Q = \left(\frac{G}{1-\mu}\right)^{1-\mu} \left(\frac{R}{\mu}\right)^{\mu}, \qquad y = Q - R.$$

Optimal internal input use preserves net output

Given G, choose R to maximize net output $Q(G,R) - R$:

$$\frac{\partial Q}{\partial R} = \frac{\mu Q}{R} = 1 \quad \Longrightarrow \quad R = \mu Q.$$

Substituting into the production function gives

$$Q = \frac{G}{1-\mu}, \qquad R = \frac{\mu G}{1-\mu}, \qquad \boxed{y = G.}$$

- The optimal ratio $R/Q = \mu$ is the same in every state; the quantities vary with G.
- Every concave CRS composite G admits this domestic roundabout representation.

Timing and accounting | General IO networks | Back to model | Appendix guide

Roundabout production: timing and resource accounting

Timing

Technology: μ is a fixed parameter.

Before uncertainty: choose external input quantities and processing labor.

After uncertainty: deliveries determine G; optimal internal production implies

$$Q = G/(1-\mu), \qquad R = \mu Q.$$

Internal quantities vary with G; external orders and processing labor remain fixed after shocks.

Gross versus net accounting

At a common unit value p,

$$pQ = pG + pR, \qquad p(Q-R) = pG.$$

The internal transaction pR circulates within the stage.

The country resource constraint counts external orders and labor once.

Economic interpretation: a static input–output loop creates intermediate transactions while leaving net output unchanged. Richer feedback can change the net technology.

General domestic networks | Back to model | Appendix guide

Replacing the sequence with a domestic IO network

For each activity v, r_v contains delivered external inputs and labor chosen ex ante. Internal flows may form cycles. After deliveries, choose gross output q, internal use x, and final use c:

$$H_i(r) = \max_{q,x,c} \; h_i(c) \quad \text{subject to} \quad \begin{cases} q_v \le f_v(r_v, \{x_{uv}\}_u), \\ c_v + \sum_w x_{vw} \le q_v, \qquad q, x, c \ge 0. \end{cases}$$

Here x_{uv} flows from activity u to v; h_i aggregates final uses into the destination's composite.

Sufficient conditions for a concave CRS net technology

Nondecreasing, concave CRS technologies f_v and final aggregator h_i.
Linear material balance; an attained optimum with finite, positive net output.
Scaling feasible plans gives CRS; combining them gives concavity of H_i.

The same sourcing condition: with log utility, predetermined prices and budget, and an interior regular optimum,

$$\frac{p_k m_k}{B_i^{\text{tot}}} = \mathbb{E}\left[\frac{\partial \ln H_i}{\partial \ln m_k}\right], \qquad B_i^{\text{tot}} = \sum_k p_k m_k.$$

m_k ranges over external input quantities and processing labor chosen ex ante; deliveries equal $\chi_k m_k$ (labor has $\chi_k = 1$). Internal transactions are not added to this resource budget.

Timing, limits, and estimation | Cobb–Douglas benchmark | Back to model

Roundabout IO: timing, limits, and estimation

What is fixed before shocks?

- Orders by origin, destination, and activity; labor by activity.
- Ex ante prices and budget; no additional purchases after shocks.
- Technology, network links, and the information used to choose orders.

After delivery: efficient internal routing and final-use allocation may respond to realized output.

What does the reduction establish?

- The sourcing condition survives; propagation must be rederived.
- To reproduce H_i, predetermined internal allocation rules must be efficient in every state.
- Import-dependent exporters require a global feasibility and delivery model.

Quantitative challenge: identify heterogeneous substitution across industries. IO shares alone do not identify elasticities.

One approach to flexible substitution

Adão, Costinot, and Donaldson (AER, 2017) estimate a *mixed-CES reduced factor-demand system*, using random coefficients to allow richer substitution among countries' factor services. Adapting this approach to sectoral input choices made before shocks requires additional identification.

Nonparametric Counterfactual Predictions in Neoclassical Models of International Trade, Section V.A, equation (28), pp. 658–661.

Network setup | Cobb–Douglas benchmark | Back to model | Appendix guide

Cobb–Douglas networks: when are elasticities invariant to risk?

Cobb–Douglas down to each individual input

Suppose every activity and final-use aggregator is Cobb–Douglas, with fixed exponents and a productive IO network. The efficient net technology is

$$H_i(m, \chi) = K_i(\chi) \prod_k m_k^{\theta_{ik}}, \qquad \sum_k \theta_{ik} = 1.$$

m_k includes each bilateral order and each labor task. The network determines the constant exponents θ_{ik}.

$$\boxed{\mathcal{E}_{ik}(\chi) = \theta_{ik} \quad \Rightarrow \quad \mathbb{E}[\mathcal{E}_{ik}] = \theta_{ik}.}$$

Changing delivery-risk variances or correlations leaves these elasticities unchanged.

Sector Cobb–Douglas with CES origin sourcing

If instead the reduced technology is Cobb–Douglas in sector bundles $B_{i,s}$,

$$H_i = K_i(\ell^p) \prod_s B_{i,s}^{\theta^B_{i,s}},$$

and those bundles retain our CES origin nests, then

$$\mathcal{E}_{ji,s}(\chi) = \theta^B_{i,s}\, \omega_{ji,s}(\chi).$$

The sector-composite elasticity $\theta^B_{i,s}$ is fixed; the bilateral elasticity can respond to risk through the within-sector share $\omega_{ji,s}$.

Implication: rich IO links alone do not create risk-sensitive elasticities in a fully Cobb–Douglas network. Non-Cobb–Douglas substitution can restore that channel.

Fixed exponents and links; exogenous multiplicative shocks. Risk can still affect output and welfare when expected elasticities are constant.

Network setup | Timing and estimation | Back to pointwise elasticity | Appendix guide

Covariance: separable country and stage correlations

For standardized latent Gaussian shocks $Z_{j,s}$, ordered by country then stage,

$$\mathrm{Corr}(Z_{j,s}, Z_{j',s'}) = R^J_{jj'} R^S_{ss'}, \qquad \Sigma_{\log \chi} = D_\sigma (R^J \otimes R^S) D_\sigma.$$

Here R^J and R^S are the country and stage correlation matrices; $\otimes$ is the Kronecker product. $D_\sigma = \mathrm{diag}(\sigma_{j,s})$ contains log-shock standard deviations. These are correlations of log shocks, not delivery levels.

Flexible marginal risk

Every country-stage pair has its own $\sigma_{j,s}$.

For 50 countries and 3 stages: 150 volatilities, 1,225 country correlations, and 3 stage correlations.

Restricted comovement

The country-correlation pattern is common across stages; the stage-correlation pattern is common across countries.

Cross-country, cross-stage correlation is their product.

Economic restriction: country and stage comovement must factor into this product, although both correlation matrices may be dense.

Estimation pools 27 annual observations and uses shrinkage after removing common stage-year movements. The implied hedging opportunities depend on this covariance structure.

Back to identification Appendix guide

Lognormal delivery risk and tail behavior

$$\chi = \exp\left(-\tfrac{1}{2}\sigma^2 + \sigma Z\right), \qquad Z \sim N(0,1).$$

What is held constant?

$$\mathbb{E}[\chi] = 1, \qquad \mathrm{Var}(\chi) = e^{\sigma^2} - 1, \qquad \mathrm{median}(\chi) = e^{-\sigma^2/2}.$$

σ^2 is log variance. At fixed orders, expected deliveries stay fixed; expected final output need not.

- χ scales deliveries relative to their expectation: favorable realizations may exceed one.
- This is a delivery-efficiency factor: it is strictly positive and has an unbounded upper tail.
- Expected utility depends on the full shock distribution, including adverse joint realizations. Equal variances and correlations need not imply equal welfare under different tail behavior.

Back to estimated risk Appendix guide

CRRA preferences: labor and input expenditure

Index each bilateral order or processing task by k; its quantity x_k is chosen ex ante and its price is p_k. At fixed ex ante prices and income Y_i, choose all inputs and final-good labor:

$$\max_{\ell_i^F, x} \mathbb{E}[u_\gamma(C_i)], \qquad C_i = (\ell_i^F)^\beta f_i(x,\chi)^{1-\beta}, \qquad w_i \ell_i^F + \sum_k p_k x_k \le Y_i,$$

Here $C_i = q_i$ is consumption and $f_i(x,\chi) = M_i$ is the CRS input composite. Relative risk aversion is $\gamma > 0$, so $u'_\gamma(C) = C^{-\gamma}$. Let ξ_i be the ex ante budget multiplier and $\mathcal{E}_{ik} = \partial \log f_i / \partial \log x_k$.

Interior first-order conditions and Euler's theorem

$$\xi_i w_i \ell_i^F = \beta \mathbb{E}[C_i^{1-\gamma}], \qquad \xi_i p_k x_k = (1-\beta)\mathbb{E}[C_i^{1-\gamma}\mathcal{E}_{ik}].$$

Since $\sum_k \mathcal{E}_{ik} = 1$,

$$\xi_i Y_i = \mathbb{E}[C_i^{1-\gamma}], \qquad w_i \ell_i^F = \beta Y_i, \qquad B_i^{\text{tot}} := \sum_k p_k x_k = (1-\beta) Y_i.$$

Thus $\ell_i^F = \beta L_i$ also under CRRA when $Y_i = w_i L_i$.

Sourcing shares | Back to allocation | Appendix guide

CRRA sourcing shares weight states by consumption

For any order or processing task k, let x_k be its quantity, p_k its price, and $\mathcal{E}_{ik} = \partial \log M_i / \partial \log x_k$ its realized output elasticity. The full supply-chain budget is $B_i^{\text{tot}} = (1-\beta)Y_i$.

The normalized expenditure-share condition

$$s_{ik}^{\text{task}} := \frac{p_k x_k}{B_i^{\text{tot}}} = \frac{\mathbb{E}[C_i^{1-\gamma}\mathcal{E}_{ik}]}{\mathbb{E}[C_i^{1-\gamma}]} = \mathbb{E}[\varpi_i(\chi)\mathcal{E}_{ik}], \qquad \varpi_i := \frac{C_i^{1-\gamma}}{\mathbb{E}[C_i^{1-\gamma}]}.$$

- **Log utility,** $\gamma = 1$**:** $\varpi_i = 1$ and $s_{ik}^{\text{task}} = \mathbb{E}[\mathcal{E}_{ik}]$.
- $\gamma > 1$**:** low-consumption states receive more weight.
- $0 < \gamma < 1$**:** high-consumption states receive more weight in this elasticity formula; utility remains concave.

The state weight satisfies $\mathbb{E}[\varpi_i] = 1$ and is proportional to $C_i u'_\gamma(C_i)$: an output elasticity measures a *proportional* consumption gain.

The allocation condition holds at given ex ante input prices and nominal income.

Back to allocation Appendix guide

Nominal and real valuation of risk

Let $P_i(\chi)$ be the consumption price and $X_i(\chi) = P_i(\chi)C_i(\chi)$ nominal consumption expenditure. Unlike $X_{ji,s}$, X_i is not a bilateral trade flow.

One extra unit of consumption

$$m_i^R(\chi) \;\propto\; u'_\gamma(C_i) = C_i^{-\gamma}.$$

Under log utility: $m_i^R \propto 1/C_i$.

Real marginal utility rises in bad consumption states.

One extra nominal dollar

$$m_i^N(\chi) \;\propto\; \frac{u'_\gamma(C_i)}{P_i} = \frac{C_i^{1-\gamma}}{X_i}.$$

Under log utility: $m_i^N \propto 1/X_i$.

A dollar buys fewer goods when the consumption price is high.

With log utility and fixed $X_i = Y_i$, $P_i = Y_i/C_i$: an extra physical unit earns more revenue in low-consumption states. Prices transmit the value of scarce consumption to firms.

Kernels are stated up to a positive date-0 scalar; state prices also include state probabilities.

When is a kernel constant? Back to allocation Appendix guide

Deterministic nominal income: constant total cash flow

Fix one nominal unit across states. With labor income, realized aggregate profits, net transfers, and net asset payoffs,

$$Y_i(\chi) = w_i(\chi)L_i(\chi) + \Pi_i(\chi) + T_i(\chi) + A_i(\chi).$$

Necessary and sufficient: the total cash flow is constant

$$Y_i(\chi) = \bar{Y}_i \quad \text{a.s.} \quad \Longleftrightarrow \quad w_i L_i + \Pi_i + T_i + A_i = \bar{Y}_i \quad \text{a.s.}$$

Here "a.s." means with probability one. With finite second moments, equivalently $\mathrm{Var}(Y_i) = 0$.

- Fixed wage payments, zero realized aggregate profits, and fixed net transfers/payoffs are **sufficient**.
- Each component need not be fixed: state-contingent payoffs can offset other income movements.
- Zero *expected* profits does not mean zero realized profits; fixed wage income alone is insufficient.

With no saving or borrowing, consumption expenditure is $X_i = P_i C_i = Y_i$. Constant nominal income and a constant nominal pricing kernel are separate conditions.

Log versus general CRRA | Back to allocation | Appendix guide

When is the nominal pricing kernel constant?

For positive consumption C_i and nominal expenditure $X_i = P_i C_i$, with CRRA coefficient γ,

$$m_i^N \propto \frac{C_i^{1-\gamma}}{X_i} \quad \Longrightarrow \quad \boxed{m_i^N \text{ constant} \iff X_i = \kappa_i C_i^{1-\gamma} \text{ a.s.}, \quad \kappa_i > 0.}$$

Log utility

$\gamma = 1$ implies

$$m_i^N \text{ constant} \iff X_i \text{ constant.}$$

If $X_i = Y_i = w_i L_i$ is fixed, the nominal kernel is constant even though real consumption and $1/C_i$ fluctuate.

General CRRA

For $\gamma \neq 1$ and fixed X_i,

$$m_i^N \text{ constant} \iff C_i \text{ constant.}$$

Conversely, a constant nominal kernel can coexist with stochastic expenditure if $X_i \propto C_i^{1-\gamma}$.

Financial insurance: payoffs affect welfare through feasible consumption transfers. In the baseline, input quantities and labor are chosen before shocks, with no contingent consumption transfers.

Back to allocation Appendix guide

Stage groups and input–output links

Industries are grouped by upstreamness. The ordering summarizes average position in the production process; it does not require every transaction to move forward.

Classification of intermediate links

Supplier \ user	Stage 1	Stage 2	Stage 3
Stage 1	Within	Forward	Forward
Stage 2	Backward	Within	Forward
Stage 3	Backward	Backward	Within

Three types of links

Forward: inputs flow to a later group.

Within: both industries belong to the same group.

Backward: inputs flow to an earlier group.

- The sequential model describes a **net assembly technology**; the empirical stage groups summarize industry upstreamness.
- Within-group intermediate use can be consistent with that net technology, as the roundabout example illustrates.
- Feedback across groups changes the network structure; an upstreamness ranking alone does not rule it out.

Roundabout production

Back to data Appendix guide

Precision of the cross-stage parameter

Let $b^M(\eta, \alpha(\eta))$ collect the three model tariff-response coefficients, with stage weights $\alpha(\eta)$ fitted to expenditure levels at each η. Let $\widehat{V}$ be the covariance matrix of the corresponding empirical coefficients.

Conditional local approximation

$$\mathbf{g}_\eta = \left.\frac{db^M(\eta, \alpha(\eta))}{d\eta}\right|_{\widehat{\eta}}, \qquad \widehat{\mathrm{se}}(\widehat{\eta}) = \left(\mathbf{g}_\eta' \widehat{V}^{-1} \mathbf{g}_\eta\right)^{-1/2}.$$

$$\widehat{\eta} = -0.74, \qquad \text{approximate conditional SE} = 0.433.$$

- Empirical coefficient covariance is clustered by importer–stage.
- The derivative uses the model coefficients at $\eta = -0.77$ and -0.71; one-sided approximations give SEs of 0.441 and 0.426.
- This calculation holds lower-nest parameters and the shock process fixed. It excludes uncertainty in those estimates and simulation error.

Estimation panel: 50 countries, 3 stages, 27 years (1995–2021).

Objective profile | Back to estimates | Appendix guide

How sharply is cross-stage substitution identified?

The minimum-distance objective $\mathcal{Q}(\eta)$ measures the covariance-weighted gap between model and empirical tariff-response coefficients.

Candidate η	$\mathcal{Q}(\eta) - \mathcal{Q}(-0.74)$
-0.95	0.178
-0.74	0
-0.71	0.007
0.20	21.927
0.50	96.177

Limited precision among negative values

All nine evaluated negative values, -0.95 through -0.71, have $\Delta\mathcal{Q} < 3.84$, the conventional one-parameter 95% cut-off.

The evaluated grid does not locate the endpoints of a confidence set.

- Estimation selects among negative values. The positive candidates are additional fit comparisons, outside that selection region.
- These comparisons do not establish rejection of Cobb–Douglas: $\eta = 0$ is not evaluated here.
- Both the profile and the local SE are conditional on the other estimated parameters.

Conditional SE · Back to estimates · Appendix guide

China: two channels of specialization

Let j denote China and superscript 0 the deterministic benchmark. Raise independent mean-one lognormal risk by $d\sigma^2_{ji,s} = d\nu_j$ for all i, s. Hold wages, prices, and processing labor fixed; orders adjust.

$$T_{i,s} := \sum_k s_{ki,s} = \frac{\sum_k X_{ki,s}}{(1-\beta)w_i L_i}, \qquad \pi_{ji,s} := \frac{s_{ji,s}}{T_{i,s}}, \qquad v^0_{i|j,s} := \frac{X^0_{ji,s}}{\sum_h X^0_{jh,s}}.$$

T includes processing in its denominator; π is China's within-stage share; v^0 weights destinations.

$$D_s := -\sum_i v^0_{i|j,s} \frac{d\log \pi_{ji,s}}{d\nu_j}, \qquad B_s := \sum_i v^0_{i|j,s} \frac{d\log T_{i,s}}{d\nu_j}.$$

Downstream relative to upstream production labor

$$\frac{d\log\left(N^M_{j,3}/N^M_{j,1}\right)}{d\nu_j} = \underbrace{D_1 - D_3}_{\text{relative within-stage sourcing response}} + \underbrace{B_3 - B_1}_{\text{relative stage-budget response}}.$$

- A larger within-stage loss upstream raises the downstream-to-upstream labor ratio.
- A larger expenditure shift toward downstream stages reinforces that effect.

$N^M_{j,s}$ is order-production labor, excluding processing labor. All derivatives are local.

Finite GE decomposition

Back to China Appendix guide

Interpreting the comparative statics

- **A mean-preserving increase in risk.** The lognormal delivery factor has $\mathbb{E}[\chi] = 1$ as dispersion changes. At fixed orders, mean deliveries are unchanged; mean final output need not be.
- **Position versus stage characteristics.** With $\psi_s = 1$ and identical stages, the technology is invariant to relabeling stage positions. Differences in ρ_s can still generate different responses across stage groups.
- **Within-stage substitution.** The elasticity is $1/(1 - \rho_s)$: a higher ρ_s makes alternative origins closer substitutes.
- **Processing labor and propagation.** When $\psi_s < 1$, inherited output combines with processing labor before entering the next stage. This makes the position of a shock relevant to its downstream effect.
- **Equilibrium adjustment.** The figures allow wages and sourcing to adjust. Fixed-price formulas isolate particular channels and need not predict the full equilibrium response.

Back to comparative statics Appendix guide

Aggregating the welfare cost of risk

Moving from the deterministic economy to the risky baseline,

$$\Delta U_i = \mathbb{E} \log C_i^{\text{risky}} - \log C_i^{\text{det}}, \qquad \text{CE}_i = \exp(\Delta U_i) - 1.$$

The consumption equivalent CE_i is the proportional change in deterministic consumption that delivers the risky economy's expected utility.

Summary across countries	Change
Unweighted mean of $100\Delta U_i$	-13.84 log points
Risky-income-weighted mean $100\overline{\Delta U}$	-15.15 log points
Equivalent uniform consumption change $e^{\overline{\Delta U}} - 1$	-14.06%

Global consumption equivalent: a uniform consumption reduction of 14.06% generates the same income-weighted welfare loss.

Weights are proportional to baseline risky-economy $w_i L_i$. Log points are $100\Delta U_i$: thus $\overline{\Delta U} = -0.1515$ in the exponential. Averaging country consumption equivalents with these same weights gives a different statistic (-12.90%).

◂ Risky vs. deterministic

◂ Appendix guide

China's shift toward downstream production

In the China risk counterfactual, the ratio of stage-3 to stage-1 order-production labor rises by 9.30% (8.90 log points).

Contribution to the relative employment change	Log points
Relative change in China's within-stage sourcing shares	$+9.37$
Relative change in destinations' stage-budget shares	$+0.21$
Destination-income composition	-0.68
Total $\Delta \log\left(N^{M}_{\mathrm{CHN},3}/N^{M}_{\mathrm{CHN},1}\right)$	$+8.90$

Main force: the relative change in China's sourcing shares favors downstream production. Wages and destination spending adjust in general equilibrium.

$T_{i,s}$ is stage-s input spending divided by the full supply-chain budget; $\pi_{\mathrm{CHN},i,s}$ is China's share of that stage's input spending. Each contribution is downstream minus upstream. Exact accounting uses log-mean destination-sales weights (Sato–Vartia) and $X_{\mathrm{CHN},i,s} = [(1-\beta)w_i L_i]T_{i,s}\pi_{\mathrm{CHN},i,s}$. The within-stage contribution includes endogenous prices. This decomposition describes the finite equilibrium change.

◂ China counterfactual Local specialization mechanism ◂ Appendix guide

China: lower input prices cushion the welfare loss

Higher risk lowers China's wage and the price of inputs produced in China.

Object	Change
China wage and China-origin input prices	-8.00%
China expected log consumption	-3.96 log points
China consumption-equivalent welfare	-3.88%
Bought-input-price contribution to China's welfare	$+1.34$ log points
Processing-wage contribution to China's welfare	$+0.63$ log points

Price channel: cheaper inputs partly offset the losses from lower income and greater production risk.

Prices use the same numeraire in both economies. Price changes are percentages; contributions to expected log consumption are log points. The total consumption equivalent is $100[\exp(-0.0396) - 1] \simeq -3.88\%$.

◂ China welfare

◂ Appendix guide

Estimation: sourcing patterns and stage expenditure

- **Indices:** origin j, destination i, stage s, year t. $X^{\mathrm{D}}_{ji,s,t}$ is the observed bilateral trade value.
- **Lower-nest estimation:** iterate between the model and PPML to estimate ρ_s, origin-stage competitiveness, and persistent bilateral-stage wedges. Gravity coefficients are denoted $\boldsymbol{b}^{\mathrm{grav}}$; β remains the final-production labor exponent.
- **Profile η:** hold the lower-nest objects fixed. For each candidate η, choose destination-stage weights $\alpha_{i,s}(\eta)$ to match purchased-input stage-share levels; then generate the model's time series.
- Run the same auxiliary regression in data and model. Match the coefficients on tariff-induced relative stage costs:

$$\hat{\eta} = \arg\min_{\eta \in \mathcal{G}} [\boldsymbol{b}^{\mathrm{D}} - \boldsymbol{b}^{\mathrm{M}}(\eta)]' \widehat{V}^{-1} [\boldsymbol{b}^{\mathrm{D}} - \boldsymbol{b}^{\mathrm{M}}(\eta)].$$

$\mathcal{G}$: candidate grid; $\boldsymbol{b}^{\mathrm{D}}, \boldsymbol{b}^{\mathrm{M}}$: data/model auxiliary slopes; $\widehat{V}$: covariance of the empirical slopes. Stage weights match levels; η matches responses over time.

Back

Within-stage estimation: tariffs and the risk term

Within-stage sourcing: PPML with an equilibrium-dependent offset

$$\begin{aligned}\mu_{ji,s,t} &:= \mathbb{E}[X^{\mathrm{D}}_{ji,s,t} \mid \text{regressors}] \\ &= \exp\left(\delta^{\mathrm{orig}}_{j,s,t} + \delta^{\mathrm{dest}}_{i,s,t} + \delta^{\mathrm{bil}}_{ji,s} - \phi_s \ln \tau^{\mathrm{tar}}_{ji,s,t} + o_{ji,s,t}\right), \\ o_{ji,s,t} &:= \frac{\ln \Psi_{ji,s,t}}{1-\rho_s} - \phi_s \ln w_{j,t}, \qquad \phi_s := \frac{\rho_s}{1-\rho_s}.\end{aligned}$$

- μ: conditional *mean trade value*; o: known offset within each PPML step. Ψ: model-implied risk-and-technology term, rebuilt after solving equilibrium.
- δ^{orig}, δ^{dest}, δ^{bil}: origin-stage-year, destination-stage-year, and bilateral-stage fixed effects. The destination effect absorbs the destination-income term.
- $\tau^{\mathrm{tar}}_{ji,s,t} = 1 + \mathrm{tariff}_{ji,s,t}$; the tariff coefficient identifies ϕ_s, hence $\rho_s = \phi_s/(1+\phi_s)$.
- Purchased-input stage shares are $S^h_{i,s,t} := \dfrac{\sum_j X^h_{ji,s,t}}{\sum_{j,r} X^h_{ji,r,t}}$, $h \in \{\mathrm{D}, \mathrm{M}\}$: include domestic inputs, exclude processing labor. Adjust $\alpha_{i,s}$ to match their baseline levels.

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Estimating delivery risk and its correlations

1. Construct origin-stage residuals from structural model predictions

$$r_{j,s,t} = \ln\left(\sum_i X^{\mathrm{D}}_{ji,s,t}\right) - \ln\left(\sum_i \widehat{X}^{\mathrm{M}}_{ji,s,t}\right), \qquad \widehat{X}^{\mathrm{M}}_{ji,s,t} = \widehat{p}_{ji,s,t}\widehat{M}_{ji,s,t}.$$

2. Estimate log-shock standard deviations

$$\widehat{\sigma}_{j,s} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(\widetilde{r}_{j,s,t} - \bar{\widetilde{r}}_{j,s})^2}, \qquad \bar{\widetilde{r}}_{j,s} := \frac{1}{T}\sum_t \widetilde{r}_{j,s,t}.$$

$\widetilde{r}$: residual after winsorization and removal of common stage-year movements. $\widehat{\sigma}_{j,s}$ is a standard deviation; $\widehat{\sigma}^2_{j,s}$ is a variance.

3. Fit pooled country and stage correlations

$$\Sigma^{\log\chi} = D_\sigma(R^J \otimes R^S)D_\sigma, \qquad \mathrm{Corr}(\log\chi_{j,s}, \log\chi_{j',s'}) = R^J_{jj'}R^S_{ss'}.$$

$D_\sigma = \mathrm{diag}(\sigma)$, stacking stages within each country; R^J: $J \times J$ country correlations; R^S: $S \times S$ stage correlations. Alternate updates using standardized residuals, with shrinkage toward the identity.

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Estimated correlation matrix of log delivery shocks

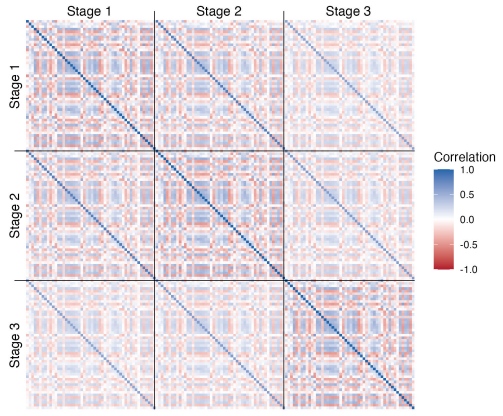


Figure: Estimated correlation matrix; rows and columns grouped by stage

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Solving the estimation fixed point

τ^{tot}: fitted trade-cost array; τ^{GE}: completed array used by the model, filling missing entries with gravity predictions.

- Complete algorithm:

Algorithm $\rho_s, a_{j,s}, \alpha_{i,s}, \boldsymbol{b}^{\text{grav}}$ estimation given η

Initialize $(\rho, a, \alpha, \boldsymbol{b}^{\text{grav}}, \ln \tau^{\text{tot}})$.
for each outer iteration **do**
    for each inner iteration **do**
        Build τ^{GE} from current fitted costs, using gravity predictions for missing entries.
        Solve GE for all t to obtain (w_t, M_t).
        Rebuild the PPML offset $o_{ji,s,t}$ from the equilibrium risk term $\Psi_{ji,s,t}$.
        Estimate gravity-like equation with PPML; extract $(\hat{\rho}, \hat{\phi}, \hat{\delta}^{\text{bil}}_{ji,s}, \hat{\delta}^{\text{orig}}_{j,s,t}, \boldsymbol{b}^{\text{grav}})$.
        Update and check inner convergence of $(\rho, a, \boldsymbol{b}^{\text{grav}}, \ln \tau^{\text{tot}})$.
    end for
    Solve GE with updated blocks and compute stage shares $S^{\text{M}}_{i,s,t}$.
    Update α to match the observed stage-share level targets.
    Check outer convergence of $(\rho, a, \alpha, \boldsymbol{b}^{\text{grav}}, \ln \tau^{\text{tot}})$.
end for

Estimation overview Back to main slides

Convergence and interpretation

PPML notation

$$\mu_{ji,s,t} := \mathbb{E}[X^{\mathrm{D}}_{ji,s,t} \mid \text{regressors}], \qquad \log \mu_{ji,s,t} : \text{ linear predictor.}$$

- With offsets and fixed-effect normalizations held fixed, the Poisson objective is concave in the linear predictor. Existence and uniqueness require the usual identification and non-separation conditions.
- Re-solving equilibrium changes the offset. The full routine is a fixed-point iteration; concavity of an individual PPML step does not establish monotone improvement of one common objective.
- Check convergence in both parameters and equilibrium objects. Global optimality is not implied.

Why PPML with lognormal shocks? Consistency uses a correctly specified conditional mean and standard regularity conditions; it does not require a Poisson conditional distribution.

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General equilibrium

Input orders and processing labor, chosen before shocks

$$p_{ji,s} M_{ji,s} = (1-\beta) w_i L_i \, \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}], \qquad p_{ji,s} = \frac{\tau_{ji,s} w_j}{a_{j,s}},$$

$$w_i \ell^p_{i,s} = (1-\beta) w_i L_i \, \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}^p_{i,s}], \qquad s = 2, \ldots, S.$$

Labor market clearing in each origin j

$$\sum_{i=1}^{J} \sum_{s=1}^{S} \frac{\tau_{ji,s} M_{ji,s}}{a_{j,s}} + \sum_{s=2}^{S} \ell^p_{j,s} = (1-\beta) L_j.$$

Final-good labor is $\ell^F_j = \beta L_j$; input production and processing use the remaining labor.

Wages, orders, and processing labor jointly satisfy these conditions, with one wage chosen as the numeraire.

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Risk adjusted productivity

- Define risk adjusted productivity:

$$\widetilde{a}_{ji,s} := \frac{a_{j,s}}{\tau_{ji,s}} \Psi_{ji,s}^{1/\rho_s} = \frac{w_j X_{ji,s}^{(1-\rho_s)/\rho_s}}{[(1-\beta) w_i L_i]^{1/\rho_s}},$$

  where $X_{ji,s} = p_{ji,s} M_{ji,s}$.

- Define the country stage risk adjusted productivity:

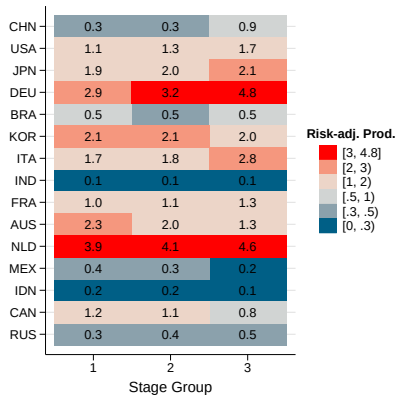
$$\widetilde{a}_{j,s} := J^{-1} \sum_i \widetilde{a}_{ji,s}$$

- Then, the risk-adjusted plotted in the heatmap is:

$$A_{j,s}^{\mathrm{RA}} := \frac{\widetilde{a}_{j,s}}{J^{-1} \sum_{j'} \widetilde{a}_{j',s}}$$

- Stage means use all origins; 1 = stage mean.

Risk-adjusted productivity, top 15 countries



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Measuring country risk centrality

- Form the $JS \times JS$ correlation matrix of the *level* delivery shocks, $C^{\chi} = \mathrm{Corr}(\boldsymbol{\chi})$, using

$$\mathrm{Corr}(\chi_{j,s}, \chi_{j',s'}) = \frac{\exp\left(\sigma_{j,s}\sigma_{j',s'}R^{J}_{jj'}R^{S}_{ss'}\right) - 1}{\sqrt{[\exp\left(\sigma^2_{j,s}\right) - 1][\exp\left(\sigma^2_{j',s'}\right) - 1]}}.$$

- Let $\boldsymbol{v}_1$ be a unit-length eigenvector associated with the largest eigenvalue of C^{χ}. Its entry $v_{1,(j,s)}$ measures the loading of origin-stage (j, s) on the dominant co-movement pattern.
- Aggregate the stage entries for country j:

$$\boxed{\mathrm{Centrality}_j = \left(\sum_{s=1}^{S} v^2_{1,(j,s)}\right)^{1/2}.}$$

- A larger value means greater participation in this common risk pattern. This is a correlation-based measure; it is distinct from the country's own log-shock standard deviations $\sigma_{j,s}$.

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Making the risk experiments comparable

Target a 5% increase in the variance of a trade-weighted *log* shock:

$$V_{\log\chi}(\varsigma^{\text{base}}) := \operatorname{Var}\left(\sum_{j,s}\varsigma^{\text{base}}_{j,s}\log\chi_{j,s}\right)$$
$$= \sum_{j,s,j',s'}\varsigma^{\text{base}}_{j,s}\varsigma^{\text{base}}_{j',s'}\sigma_{j,s}\sigma_{j',s'}R^{J}_{jj'}R^{S}_{ss'}.$$

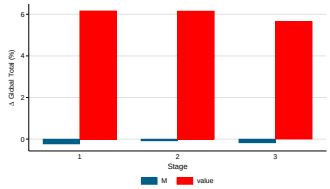
Here $\varsigma^{\text{base}}_{j,s} := \sum_i X^{\text{base}}_{ji,s} / \sum_{j',i',r} X^{\text{base}}_{j'i',r}$ is the baseline origin-stage share of world input expenditure; R^J, R^S are country and stage log-shock correlations.

- Multiply affected log variances $\sigma^2_{j,s}$ by f, hence standard deviations by $\sqrt{f}$; keep correlations fixed.
- Choose f using initial weights ς^{base}. Baseline aggregate risk is approximately 0.051; equilibrium expenditure weights may subsequently change.

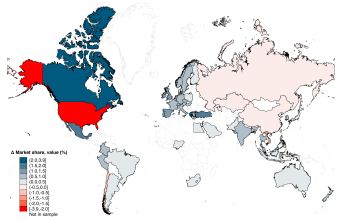
Higher risk for	Upstream	Downstream	China	US
Target increase	5%	5%	5%	5%
Log-variance scale factor f	1.30	1.23	1.10	1.30
Target aggregate risk (rounded)	0.054	0.054	0.054	0.054

Back to stage risk Back to China

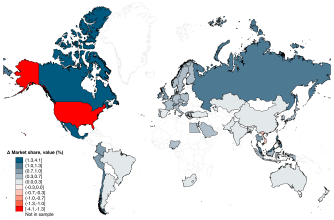
Higher US delivery risk: production and sourcing



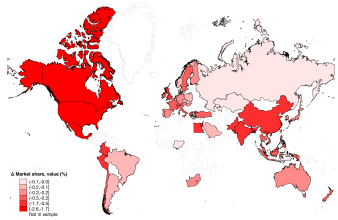
(a) Δ% Total Production



(b) Share of sales to the US: stage 1 (pp)



(c) Share of sales to the US: stage 2 (pp)



(d) Share of sales to the US: stage 3 (pp)

- Raise US log-shock variances at all stages to increase aggregate risk by 5%, using baseline weights
- World input quantities fall, while their nominal value rises
- Foreign sales to the US respond differently across stages

Back to China

Higher US delivery risk: welfare

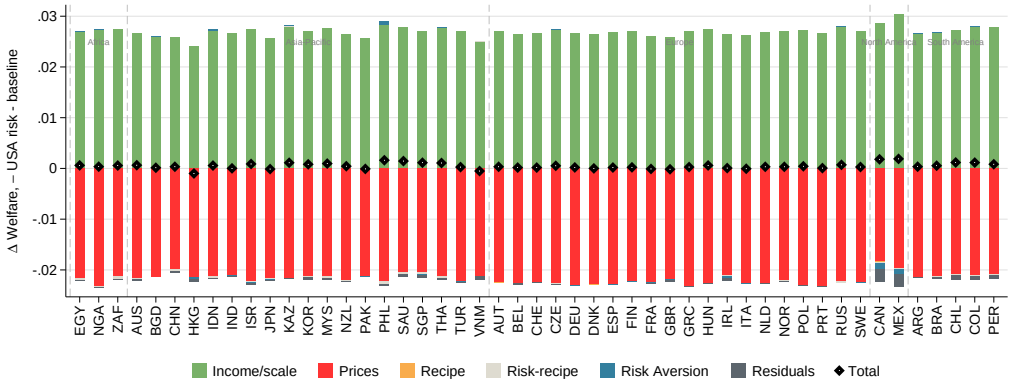
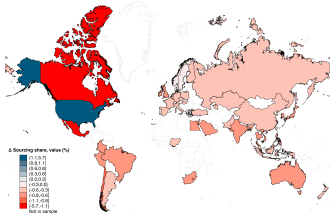


Figure: Welfare outside the US: selected contributions and total (ΔU_i)

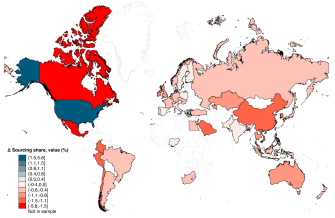
- US consumption-equivalent welfare falls by 2.35%

Back to China

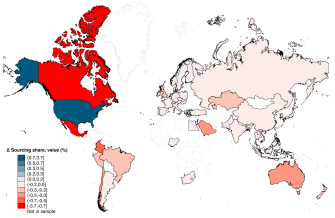
US import costs: sourcing changes by stage



(a) Sourcing from the US: stage 1 (pp)



(b) Sourcing from the US: stage 2 (pp)



(c) Sourcing from the US: stage 3 (pp)

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US import costs: welfare effects across countries

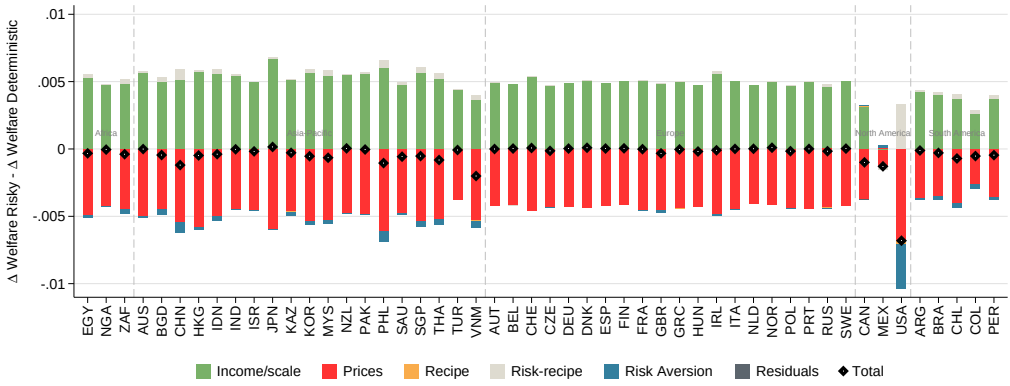


Figure: Difference in welfare responses: risky minus deterministic (log units)

- The US sources less domestically under risk, both before and after the cost increase

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US import costs: income contribution by environment

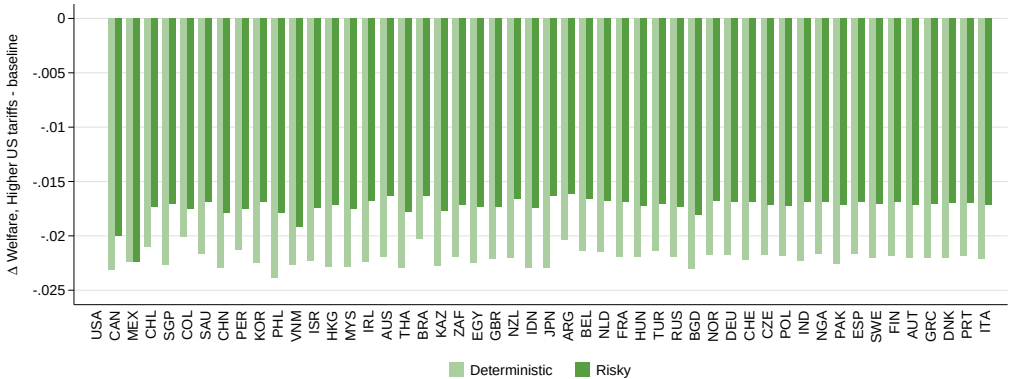


Figure: Welfare change decomposition: income, by environment

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US import costs: purchased-input price contribution

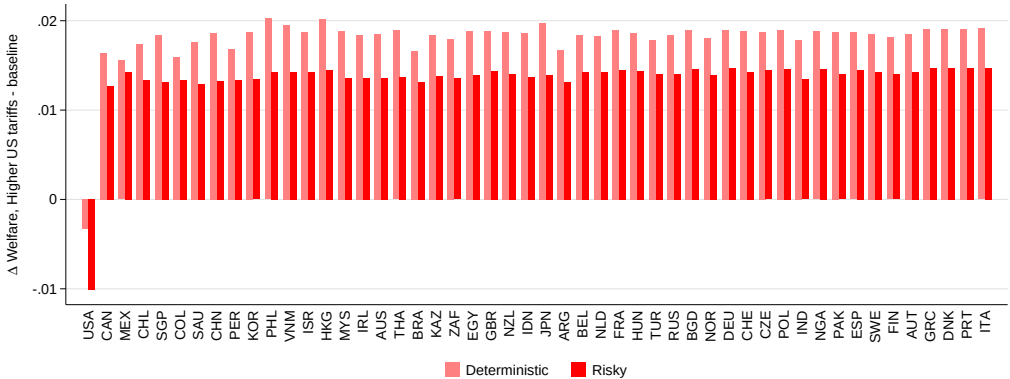
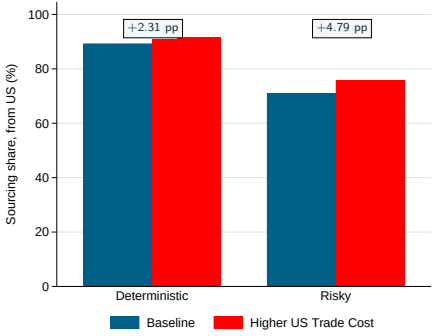


Figure: Purchased-input price contribution; processing-labor prices excluded

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US import costs: domestic sourcing responds more under risk

Figure: Change in the US domestic sourcing share



US domestic share (%)	Baseline	Higher import costs
Risky	71.0	75.8
Deterministic	89.3	91.6

- Domestic sourcing rises by 4.8 percentage points under risk, versus 2.3 points without risk

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How delivery risk changes sourcing

Risk experiment. Independent mean-one lognormal shocks: $\log \chi_{ji,s} \sim \mathcal{N}(-\sigma^2_{ji,s}/2, \sigma^2_{ji,s})$. Here $\sigma^2_{ji,s}$ is the variance of the *log* shock.

Elasticity slope and curvature. Holding all choices fixed,

$$\vartheta_{ji,s} := \frac{\partial \log \mathcal{E}_{ji,s}}{\partial \log \chi_{ji,s}}, \qquad \zeta_{ji,s} := \frac{\partial^2 \log \mathcal{E}_{ji,s}}{\partial (\log \chi_{ji,s})^2}.$$

These measure how an input's output elasticity changes with its realized delivery shock.

Direct effect on expected elasticity

How greater uncertainty changes the marginal incentive to source an input?

$$\left.\frac{\partial \mathbb{E}_{\boldsymbol{\chi}}[\mathcal{E}_{ji,s}]}{\partial \sigma^2_{ji,s}}\right|_{\mathbf{M}_i, \boldsymbol{\ell}_i^p, w, p} = \frac{1}{2}\, \mathbb{E}_{\boldsymbol{\chi}}\left[\mathcal{E}_{ji,s}(\vartheta^2_{ji,s} + \zeta_{ji,s} - \vartheta_{ji,s})\right].$$

Stein's Lemma

Order response

Explicit derivative and cascade weight